例13.

思考题：

一、(1)、$M(t)=\int_V \rho(\vec{x},t)\,dV$.

(2)、单位面积：$\dot{m}_{out}=\rho\vec{u}\cdot\hat{n}$.

$\therefore \dot{M}_{out}(t)=\iint_{\partial V}\rho(\vec{x},t)\,u(\vec{x},t)\cdot\hat{n}\,dS$.

(3)、$\frac{d}{dt}M(t)=\dot{M}_{out}(t)$.

$\therefore \frac{d}{dt}M(t)=\iint_{\partial V}\rho(\vec{x},t)\,u(\vec{x},t)\cdot\hat{n}\,dS$.

二、(1)、$\vec{P}(t)=\int_V \rho\vec{u}\,dV$.

(2)、$\vec{F}_p(t)=\iint_{\partial V}(-p\hat{n})\,dS$.

(3)、设"外力场"为单位质量所受外力.

$\therefore \vec{F}_{total}=\int_V \rho(\vec{x},t)\,\vec{F}(\vec{x},t)\,dV$.

(4)、通过面积元dS的质量流：$\rho\vec{u}\cdot\hat{n}\,dS$.

$\therefore d\vec{P}_x=-\rho u_x(\vec{u}\cdot\hat{n})\,dS$

$\therefore \vec{P}_x=-\iint_{\partial V}\rho u_x(\vec{u}\cdot\hat{n})\,dS=-\iint_{\partial V}\rho u_x\vec{u}\,d\vec{S}$

(5)、$\frac{d}{dt}\int_V \rho\vec{u}\,dV=-\iint_{\partial V}\rho\vec{u}(\vec{u}\cdot\hat{n})\,dS+\iint_{\partial V}(-p\hat{n})\,dS$
$+\int_V \rho\vec{F}\,dV$.

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书后.

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3. OA: $x=t, y=0, x+y=t, ds=1dt$

AB: $x=1-t, y=t, x+y=1, ds=\sqrt{2}dt$

BO: $x=0, y=1-t, x+y=1-t, ds=1dt$.

$\therefore$ 原式 $=\int_0^1 t\,dt+\int_0^1\sqrt{2}\,dt+\int_0^1(1-t)\,dt=1+\sqrt{2}$.

4. $x=\cos\theta\ y=\sin\theta, \theta\in[\pi,2\pi]$.

原式 $=\int_\pi^{2\pi}d\theta=\pi$.

6. 原式 $=\int_0^{2\pi}a^{2n}\cdot a\,dt=2\pi a^{2n+1}$

7. $x'(t)=-3a\cos^2t\sin t$. $y'(t)=3a\sin^2t\cos t$.

$\therefore ds=3a\cos t\sin t\,dt$.

原式 $=\int_0^{\frac{\pi}{2}}a^{\frac{4}{3}}(\cos^4t+\sin^4t)\cdot 3a\cos t\sin t\,dt$

$=3a^{\frac{7}{3}}\cdot\frac{1}{3}=a^{\frac{7}{3}}$

9. 即为圆形周长: 原式 $=4l=4\sqrt{2}$.

2. (2) $\begin{cases}x'(t)=\cos t-t\sin t\\ y'(t)=\sin t+t\cos t\\ z'(t)=1\end{cases}$

$\therefore ds=\sqrt{t^2+2}\,dt$.

$\therefore$ 原式 $=\int_0^{t_0}t\sqrt{t^2+2}\,dt=\frac{1}{3}\left[(t_0^2+2)^{\frac{3}{2}}-2\sqrt{2}\right]$

4. 解: $\int_C y\,ds$. $x=a\cos t$, $y=a\sin t$, $t\in[0,\pi]$.

$\therefore ds=a\,dt$.

$\therefore m=a^2\int_0^{\pi}\sin t\,dt=2a^2$.

5. 设 $\rho=ky$.

$\therefore \begin{cases} x'(t)=a(1-\cos t). \\ y'(t)=a\sin t. \end{cases}$

$\therefore ds=2a\sin\frac{t}{2}$.

$\therefore I=\int_C ky\cdot y^2 ds$

$=k\int_0^{2\pi}a^3(1-\cos t)^3\cdot 2a\sin\frac{t}{2}dt$

$=16ka^4\int_0^{2\pi}\sin^7\frac{t}{2}\,dt=\frac{1024}{35}a^4k$.

6. (1). $\vec{r}=(R\sin\varphi\cos\theta,\ R\sin\varphi\sin\theta,\ R\cos\varphi)$; $0\le\varphi\le\frac{\pi}{2}$

$0\le\theta\le 2\pi$

$\therefore ds=R^2\sin\varphi\,d\theta\,d\varphi$

$\therefore \iint\lambda=\int_0^{\frac{\pi}{2}}d\varphi\int_0^{2\pi}R^2\sin\varphi\cdot R\sin\varphi\sin\theta\,d\theta=0$.

(3). $\vec{r}=(R\cos\theta,\ R\sin\theta,\ z)$, $0\le\theta\le 2\pi$, $0\le z\le|-1$.

$\therefore ds=R\,d\theta\,dz$.

$\therefore \iint\lambda=\int_0^{2\pi}d\theta\cdot\int_0^{|-1}\frac{R}{R^2+z^2}dz=2\pi R\arctan\frac{|-1}{R}$.

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(4). $z_x=2x$, $z_y=2y$

$ds=\sqrt{1+4x^2+4y^2}\,dxdy$. $|xyz|=|xy(x^2+y^2)|$

取极坐标：$0\le r\le 1$, $0\le\theta\le 2\pi$

$x=r\cos\theta$, $y=r\sin\theta$

$\therefore$ 原式 $=\int_0^{2\pi}|\sin\theta\cos\theta|\,d\theta\int_0^1\sqrt{1+4r^2}\,r^5\,dr$

$=4\int_0^{\frac{\pi}{2}}\frac{1}{2}\sin 2\theta\,d\theta\int_0^1\sqrt{1+4r^2}\,r^5\,dr$

$=2\int_0^1 r^5\sqrt{1+4r^2}\,dr$

$=\frac{125\sqrt{5}-1}{420}$

7. $z_x=\frac{-x}{\sqrt{R^2-x^2-y^2}}$, $z_y=\frac{-y}{\sqrt{R^2-x^2-y^2}}$

$\therefore ds=\frac{R}{\sqrt{R^2-(x^2+y^2)}}\,dxdy$. 取 $x=r\cos\theta$, $y=r\sin\theta$

$\therefore ds=\frac{R}{\sqrt{R^2-r^2}}\,r\,dr\,d\theta$

$\therefore$ 区域：$r^2\le Rr\cos\theta$ 即 $r\le R\cos\theta$；$-\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$

$\therefore S=R\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}d\theta\int_0^{R\cos\theta}\frac{1}{\sqrt{R^2-r^2}}\,r\,dr$

$=2R^2\int_0^{\frac{\pi}{2}}(1-\sin\theta)\,d\theta=R^2(\pi-2)$.

8. 球坐标：$\sigma = r^2\sin^2\phi$

$\therefore m = \int_0^{\pi} R^4 \sin^3\phi \, d\phi \int_0^{2\pi} d\theta = \frac{8}{3}\pi R^4$

10.(4). 设 $x=t$, $y=t^3$.

$\therefore \begin{cases} x' = 1 \\ y' = 3t^2 \end{cases}$

$\therefore \cos\alpha = \frac{1}{\sqrt{9t^4+1}}$, $\cos\beta = \frac{3t^2}{\sqrt{9t^4+1}}$

$\therefore \int f\,ds = \int_C \frac{x^2y - 3x^3}{\sqrt{9x^4+1}} ds$

2. $\begin{cases} x' = 1 \\ y' = 2t \\ z' = 3t^2 \end{cases}$

$\therefore \cos\alpha = \frac{1}{\sqrt{9t^4+4t^2+1}}$, $\cos\beta = \frac{2t}{\sqrt{9t^4+4t^2+1}}$, $\cos\gamma = \frac{3t^2}{\sqrt{9t^4+4t^2+1}}$

$\int f\,ds = \int_C \frac{P + 2Qx + 3x^2R}{\sqrt{9x^4+4x^2+1}} ds.$

11.(1). 设 $x=t$, $y=t$. $t\in[0,1]$

$\therefore \vec{F} = (x^2-y^2, xy) = (0, t^2)$. $\vec{r} = (1,1)$.

$\therefore \int f\,ds = \int_0^1 \vec{F}\cdot\vec{r}\,dt = \frac{1}{3}$

2. 设 $x=t$. $y=t^2$.

$\vec{F} = (t^2-t^4, t^3)$, $\vec{r} = (1, 2t)$.

$\int f\,ds = \int_0^1 (t^2 - t^4 + 2t^4)\,dt = \frac{8}{15}$

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12 (2). $\vec{F} = (a\sin t, a\cos t)$. $\vec{r}' = (-a\sin t, a\cos t)$

$\therefore \int \vec{F} d\vec{r} = \int_0^{\frac{\pi}{4}} a^2 \cos 2t \, dt = \frac{1}{2}a^2$.

4 ∵ 设 $x = \cos t$, $y = -\sin t$, $z = 2 - \cos t - \sin t$.

$\vec{F} = (2 - \cos t, 2\cos t + \sin t - 2, -\sin t - \cos t)$

$\vec{r}' = (-\sin t, -\cos t, \sin t - \cos t)$

$\therefore \int \vec{F} d\vec{r} = \int_0^{2\pi} (-2\sin t + 2\cos t - 1) \, dt = -2\pi$.

13. $\vec{r}' = (\cos t, -\sin t, 2t)$

$\therefore W = \int_0^{\frac{\pi}{2}} (\sin^2 t \cos t - \sin^2 t \cos t + 2t^5) \, dt$

$= 2\int_0^{\frac{\pi}{2}} t^5 dt = \frac{1}{192}\pi^6$.